



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 8

Team No.: 9

Project Title: Linux Process Monitoring and Control System

Team Members

Roll Number	Student Name	Signature
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1. Abstract

The **Linux Process Monitoring and Control System** is a system-level application designed to monitor and manage processes running in a Linux operating system. The system provides users with essential information about active processes, including Process ID (PID), process name, process state, CPU utilization, memory usage, and execution details. This enables users to understand how system resources are being utilized and identify processes that may be consuming excessive resources or becoming unresponsive.

The proposed system also provides process control operations such as terminating, suspending, and resuming processes. These operations are performed using Linux process-management mechanisms and signals, demonstrating fundamental operating-system concepts such as process creation, process states, scheduling, inter-process interaction, and signal handling.

The system aims to provide a simple and organized approach to process monitoring and control while reducing the complexity of using multiple command-line utilities. It can serve as an educational tool for understanding Linux process management as well as a basic utility for observing and controlling system processes. Overall, the project demonstrates how operating-system-level mechanisms can be integrated into a practical application for efficient process monitoring, resource management, and process control.

2. Problem Statement

Managing and controlling multiple processes in a Linux environment can become challenging, especially when several applications and background services are running simultaneously. Processes continuously consume system resources such as CPU, memory, and execution time, and some processes may become unresponsive or consume excessive resources, affecting overall system performance.

Although Linux provides command-line utilities such as `ps`, `top`, and `kill` for process monitoring and control, using multiple commands and remembering their options can be inconvenient, particularly for users who are new to Linux process management. There is also a need to quickly identify important process details such as Process ID (PID), process name, current state, CPU usage, and memory usage in a structured manner.

Therefore, the proposed Linux Process Monitoring and Control System aims to provide a centralized and user-friendly mechanism for monitoring active processes and performing basic control operations. The system will allow users to identify resource-intensive processes, observe their states, and perform operations such as terminating, suspending, and resuming processes. This will make process management more organized while providing practical understanding of Linux process management, signals, and system-level operations.

3. Objectives

1. To monitor active processes running in a Linux system.
2. To display important process details such as PID, process name, state, CPU usage, and memory usage.
3. To identify processes that consume excessive system resources.
4. To provide basic process control operations such as terminate, suspend, and resume.
5. To demonstrate Linux concepts such as process states, system calls, and signals.
6. To provide a simple and organized interface for effective process monitoring and control.

4. Proposed Methodology

The **Linux Process Monitoring and Control System** will be designed as a system-level application that collects, displays, and manages information about running processes.

The major steps of the proposed methodology are:

1. **Process Detection:** Identify all active processes running in the Linux system using the `/proc` filesystem and system-level mechanisms.
2. **Information Collection:** Retrieve details such as PID, process name, process state, CPU usage, and memory usage.
3. **Process Monitoring:** Continuously update the process information to reflect changes in resource usage and process states.
4. **Process Selection:** Allow the user to select a specific process using its PID.
5. **Process Control:** Provide operations such as terminate, suspend, and resume using appropriate Linux signals.
6. **Status Update:** Refresh the process list after each operation and display the updated process status.

System Flow

Start → Detect Processes → Collect Process Information → Display Process Details → Select Process → Monitor/Control Process → Update Status → Continue Monitoring / Exit

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	Core concept of the project; manages creation, execution, monitoring, and termination of processes.
Process ID (PID)	Uniquely identifies each running process so the system can monitor or control it.
Parent-Child Processes	Helps understand process hierarchy and relationships between processes.
fork()	Creates child processes for testing and demonstrating process management.
exec()	Executes a new program/process from the application.
wait()/waitpid()	Allows the parent to monitor and collect the status of child processes.
kill()	Sends signals to processes for controlling or terminating them.
POSIX Signals	Provides process-control operations such as stop, continue, and terminate.
SIGTERM, SIGKILL, SIGSTOP, SIGCONT	Used to terminate, force-kill, pause, and resume processes.
/proc Filesystem	Retrieves real-time process information such as PID, state, CPU and memory usage.
CPU & Memory Monitoring	Monitors resource usage of individual processes.
Process Priority (nice / setpriority())	Allows monitoring and optionally modifying process scheduling priority.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030006	Bhanu Jakka	Process Monitoring Module
2520030043	Akshita Anand	Process Control Module
2520030192	A.Kuvira	Priority Management & CLI
2520030455	P. Sudha Gayathri	Integration, Testing & Documentation

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system and development environment for the project.
C / C++	Main language for implementing the process monitoring and control system.
GCC	Compiling the C source code
Linux Terminal	Code editing and debugging
Make / Makefile	Automates compilation of the project.
Git / GitHub	Version control and team Collaboration
/proc Filesystem	Provides process and system information for monitoring.

8. Expected Outcome

The Linux Process Monitoring and Control System is expected to provide a simple and efficient way to monitor and manage processes running in a Linux environment. The system will display important process information such as PID, process name, process state, CPU usage, and memory usage in an organized manner. Users will be able to identify resource-intensive processes and perform basic control operations such as terminating, suspending, and resuming processes. The system will also update the process status after each operation, providing a clear view of the current process state. Overall, the project is expected to demonstrate the practical use of Linux process management, system calls, and signals while providing an effective approach to process monitoring and control.

9. GitHub Repository

GitHub Repository Link: https://github.com/Akshita43/Cupcake_OSSP_Project

Repository Created:

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: Muniraju Vadlamudi

Signature:

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____